

ASSESSMENT TOOL

1.

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q1 Student Registration System10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2 public class StudentRegistrationSystem {
3     public static void main(String[] args) {
4         HashSet<String> students = new HashSet<>();
5         students.add("S101");
6         students.add("S102");
7         students.add("S103");
8         HashMap<String, ArrayList<String>> studentCourses = new HashMap<>();
9         studentCourses.put("S101", new ArrayList<>(Arrays.asList("Java", "Operating System")));
10        studentCourses.put("S102", new ArrayList<>(Arrays.asList("Java", "Operating System")));
11        studentCourses.put("S103", new ArrayList<>(Arrays.asList("Comp", "Operating System")));
12        System.out.println("Student Course Details");
13        System.out.println("-----");
14        Iterator<String> iterator = students.iterator();
```

Input

stdin input

Output

Student Course Details

Student ID : S102
Courses : [Java, Operating System]

Q5
Online Voting System
10 marks

5/5 answered

```
3     public static void main(String[] args) {
4         HashSet<String> students = new HashSet<>();
5         students.add("S101");
6         students.add("S102");
7         students.add("S103");
8         HashMap<String, ArrayList<String>> studentCourses = new HashMap<>();
9         studentCourses.put("S101", new ArrayList<>(Arrays.asList("Java", "Operating System")));
10        studentCourses.put("S102", new ArrayList<>(Arrays.asList("Java", "Operating System")));
11        studentCourses.put("S103", new ArrayList<>(Arrays.asList("Comp", "Operating System")));
12        System.out.println("Student Course Details");
13        System.out.println("-----");
14        Iterator<String> iterator = students.iterator();
15        while (iterator.hasNext()) {
16            String student = iterator.next();
17            System.out.println("Student ID : " + student);
18            System.out.println("Courses : " + studentCourses.get(student));
19            System.out.println();
20        }
21    }
22 }
```

Run Save

Output

Student Course Details

Student ID : S102
Courses : [Java, Operating System]

2.

QUESTIONS

Q1

Student Registration System

10 marks

Q2

E-Commerce Cart System

10 marks

Q3

Library Management System

10 marks

Q4

Employee Data Processing

10 marks

Q5

Online Voting System

10 marks

5/5 answered

Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```

1 import java.util.ArrayList;
2 import java.util.Iterator;
3 public class EcommerceCartSystem {
4     public static void main(String[] args) {
5         ArrayList<String> a = new ArrayList<>();
6         a.add("Laptop");
7         a.add("Mouse");
8         a.add("Keyboard");
9         a.add("Headphones");
10        ArrayList<String> b = new ArrayList<>();
11        Iterator<String> itr = a.iterator();
12        while(itr.hasNext()){
13            String item = itr.next();
14            if(!item.equals("mouse")){

```

Input

stdin input

Output

Items in Cart:

Laptop

Mouse

Keyboard

Headphones

Online Voting System

10 marks

5/5 answered

```

4     public static void main(String[] args) {
5         ArrayList<String> a = new ArrayList<>();
6         a.add("Laptop");
7         a.add("Mouse");
8         a.add("Keyboard");
9         a.add("Headphones");
10        ArrayList<String> b = new ArrayList<>();
11        Iterator<String> itr = a.iterator();
12        while(itr.hasNext()){
13            String item = itr.next();
14            if(!item.equals("mouse")){
15                b.add(item);
16            }
17        }
18        System.out.println("Items in Cart:");
19        Iterator<String> c = b.iterator();
20        while (c.hasNext()) {
21            System.out.println(c.next());
22        }
23    }
24 }

```

Run Save

Output

Items in Cart:

Laptop

Mouse

Keyboard

Headphones

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3.

Q1 Student Registration System
10 marks

Q2 E-Commerce Cart System
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Q3 Library Management System
10 marks

Q4 Employee Data Processing
10 marks

Q5 Online Voting System
10 marks

5/5 answered

Q3 Library Management System 10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```

1 import java.util.HashMap;
2 import java.util.Map;
3 public class LibraryManagementSystem {
4     public static void main(String[] args) {
5         HashMap<String, String> books = new HashMap<>();
6         books.put("978001", "Java Programming");
7         books.put("978002", "Data Structures");
8         books.put("978003", "Operating Systems");
9         System.out.println("Book with ISBN 978002:");
10        System.out.println(books.get("978002"));
11        System.out.println("\nAll Books:");
12        for (Map.Entry<String, String> entry : books.entrySet()) {
13            System.out.println(entry.getKey() + " -> " + entry.getValue());
14        }
15    }
                
```

Input

stdin input

Output

```

Book with ISBN 978002:
Data Structures

All Books:
978001 -> Java Programming
978002 -> Data Structures
978003 -> Operating Systems
                
```

4.

Q1 Student Registration System
10 marks

Q2 E-Commerce Cart System
10 marks

Q3 Library Management System
10 marks

Q4 Employee Data Processing
10 marks

Q5 Online Voting System
10 marks

5/5 answered

Q4 Employee Data Processing 10 marks

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```

1 import java.util.Iterator;
2 import java.util.ArrayList;
3 import java.util.List;
4 public class Employee {
5     public static void main(String[] args) {
6         List<EmployeeDetails> employees = new ArrayList<>();
7         employees.add(new EmployeeDetails(1, "Alice", 45000));
8         employees.add(new EmployeeDetails(2, "Bob", 55000));
9         employees.add(new EmployeeDetails(3, "Carol", 72000));
10        employees.add(new EmployeeDetails(4, "David", 30000));
11        Iterator<EmployeeDetails> it = employees.iterator();
12        System.out.println("Employees with salary > 50000:");
13        while (it.hasNext()) {
14            EmployeeDetails e = it.next();
15            if (e.getSalary() > 50000) {
                
```

Input

stdin input

Output

```

Employees with salary > 50000:
Employee(empId=2, name='Bob',
salary=55000.0)
Employee(empId=3, name='Carol',
salary=72000.0)
                
```

5/5 answered

```

8     employees.add(new EmployeeDetails(2, "Bob", 55000));
9     employees.add(new EmployeeDetails(3, "Carol", 72000));
10    employees.add(new EmployeeDetails(4, "David", 30000));
11    Iterator<EmployeeDetails> it = employees.iterator();
12    System.out.println("Employees with salary > 50000:");
13    while (it.hasNext()) {
14        EmployeeDetails e = it.next();
15        if (e.getSalary() > 50000) {
16            System.out.println(e);
17        }
18    }
19 }
20 }
21 class EmployeeDetails {
22     private int empId;
23     private String name;
24     private double salary;
25     public EmployeeDetails(int empId, String name, double salary) {
26         this.empId = empId;
27         this.name = name;
28         this.salary = salary;
29     }
30     public double getSalary() {
31         return salary;
32     }
33     public String toString() {
34         return "Employee(empId=" + empId + ", name='" + name + "', salary=" + salary + ")";
35     }
36 }

```

```

Employees with salary > 50000:
Employee(empId=2, name='Bob',
salary=55000.0)
Employee(empId=3, name='Carol',
salary=72000.0)

```

5.

QUESTIONS

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Q5
Online Voting System
10 marks

5/5 answered

Q5 Online Voting System 10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting
- Display results

Your Code

```

1 import java.util.HashMap;
2 import java.util.HashSet;
3 import java.util.Map;
4 public class OnlineVotingSystem {
5     public static void main(String[] args) {
6         HashSet<String> voters = new HashSet<>();
7         HashMap<String, Integer> votes = new HashMap<>();
8         votes.put("Alice", 0);
9         votes.put("Bob", 0);
10        castVote("V101", "Alice", voters, votes);
11        castVote("V102", "Bob", voters, votes);
12        castVote("V103", "Alice", voters, votes);
13        castVote("V101", "Bob", voters, votes);
14        System.out.println("\nVoting Results");

```

Input

stdin input

Output

```

V101 voted for Alice
V102 voted for Bob
V103 voted for Alice
Duplicate vote rejected for V101

```

Q5

Online Voting System
10 marks

5/5 answered

```
4 public class OnlineVotingSystem {
5     public static void main(String[] args) {
6         HashSet<String> voters = new HashSet<>();
7         HashMap<String, Integer> votes = new HashMap<>();
8         votes.put("Alice", 0);
9         votes.put("Bob", 0);
10        castVote("V101", "Alice", voters, votes);
11        castVote("V102", "Bob", voters, votes);
12        castVote("V103", "Alice", voters, votes);
13        castVote("V101", "Bob", voters, votes);
14        System.out.println("\nVoting Results");
15        for (Map.Entry<String, Integer> entry : votes.entrySet()) {
16            System.out.println(entry.getKey() + " : " + entry.getValue());
17        }
18    }
19    public static void castVote(String voterId, String candidate,
20                                HashSet<String> voters,
21                                HashMap<String, Integer> votes) {
22        if (voters.contains(voterId)) {
23            System.out.println("Duplicate vote rejected for " + voterId);
24        } else {
25            voters.add(voterId);
26            votes.put(candidate, votes.get(candidate) + 1);
27            System.out.println(voterId + " voted for " + candidate);
28        }
29    }
30 }
```

Output

```
V101 voted for Alice
V102 voted for Bob
V103 voted for Alice
Duplicate vote rejected for V101
```

Voting Results